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A
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on
An Intelligent Alumni Management System with Machine Learning-Based Placement Prediction and Predictive Analytics

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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

We also declare that the work presented in this report is original, authentic, and has been carried out independently under the supervision of Mr. Tanmoy Das, Assistant Professor, Department of Computer Science and Engineering (AI&ML), KIET Group of Institutions, Ghaziabad. The information and data used in this project have been collected from reliable and authentic sources, and due credit has been given wherever required. We confirm that this work has not been plagiarized, and all references have been appropriately cited.

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CERTIFICATE

This is to certify that Project Report entitled "Alumni Management System" which is submitted by Tanishq Kumar, Tarang Sharma, Vidit Raj Jain in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Computer Science and Engineering (Artificial intelligence) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ABSTRACT

Alumni management systems (AMS) are essential for many schools and universities to maintain databases and facilitate interaction and networking among alumni. These networks are vital to the development of the university and are used for mentoring programs, cooperation with industries, and information exchange. However, existing AMS solutions tend to be simple data warehouses where users store information about alumni, send messages and organize meetings or conferences but do not provide much data-driven intelligence in form of analysis or any other form of useful insights into student achievements, employability trends and placements. The era of academic competition makes it crucial for educational institutions to use advanced systems that can analyse student progress and provide useful recommendations to improve their employability.

The goal of this study was to design and implement an Intelligent Alumni Management System based on placement prediction and predictive analytics through the use of machine learning. This system was created in the form of a web application using Spring Boot as the backend, JSP technology on the front end and MySQL database for storing data. A supervised machine learning algorithm processes information about students' performance in college and predicts the probability of being hired after graduation. Four main classifiers were used and evaluated by accuracy, precision, recall, and F1-score metrics: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM). To start the prediction, data are pre-processed using cleaning, feature selection, encoding categorical features, normalizing and train-test split.

According to results, random forests outperformed all other classifiers and provided stable predictions with the highest accuracy. Incorporation of predictive analytics into the alumni management system allows identifying students with different degrees of employment probabilities and helps organizations develop relevant training and mentoring programs to improve their outcomes. Thus, the use of machine learning algorithms significantly improves functionality of existing AMSs providing a number of advantages for their operation.

This new approach addresses the existing void between conventional alumni record-keeping and intelligent, data-driven institution management through the integration of machine learning in the alumni management tool. The system is capable of providing real-time placement prediction, proactive academic intervention, and analytical dashboards for management personnel.

Keywords: Alumni Management System, Machine Learning, Placement Prediction, Predictive Analytics, Random Forest, Decision Tree, Logistic Regression, Support Vector Machine, Educational Data Mining, Student Performance Analysis.

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LIST OF ABBREVIATIONS

NAM	Network Animator
AMS	Alumni Management System
ML	Machine Learning
AI	Artificial Intelligence
SVM	Support Vector Machine
RF	Random Forest
DT	Decision Tree
LR	Logistic Regression
JSP	Java Server Pages
DBMS	Database Management System
API	Application Programming Interface
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
EDM	Educational Data Mini

CHAPTER 1

INTRODUCTION

1.1 INRODUCTION

Currently, alumni engagement and placement performances have gained considerable attention for evaluating the efficiency of an educational organization. It is because alumnus helps to provide support through mentoring, network building, collaboration in industries, and exchange of knowledge. Meanwhile, student placements are critical as they determine the career opportunities and influence the standing and prestige of the institution. As a result, efficient alumni interaction and enhanced placement are among significant goals of modern educational systems.

The current generation of Alumni Management Systems (AMS) performs only basic tasks associated with collecting, storing, managing, and processing communication activities and events. The typical examples include alumni registration and records maintenance, alumni event management, etc. These AMS are generally built using centralized databases. Such systems provide great functionality but do not offer any advanced analytics possibilities. Specifically, most of the traditional systems cannot provide valuable insights into student performance, their employability, and their placement probability. Therefore, educational institutions use manual analysis, which can be time-consuming and inaccurate.

On the other hand, recent advancements in Machine Learning (ML) and Educational Data Mining have opened up new possibilities for effective data analysis, including prediction of student performance [6], [7]. Machine learning makes it possible to extract valuable patterns and relationships in huge amounts of data that are invisible by conventional means of analysis. With this technology, the institution can evaluate numerous criteria (such as academic results, technical and professional skills, internships, projects, communication skills, etc.) to assess the possibility of placement.

It is worth noting that the combination of big data, cloud computing, and machine learning algorithms has changed the way educational organizations can engage with their data. In contrast to former approaches that were limited due to the computing power and absence of adequate algorithms, current organizations now have access to a great number of open-source solutions that make it possible to use machine learning in even low-resource environments. The presence of libraries like Scikit-learn, Pandas, and NumPy has made complex algorithms accessible for everyone, and even small groups of developers can develop solutions that would require full-scale data science departments formerly.

One can apply several different supervised machine learning algorithms (e.g., Logistic

Regression, Decision Tree, Random Forest, SVM) to predict student placement probability. These models will help the institution classify the students according to their probability of getting placements and identify those students who would make valuable contributions to general placement statistics. Moreover, the model will be able to point at students who need to improve specific skills or acquire more training to become more employable [1], [4].

For higher education institutions in India, placements statistics have proved to be a significant measure of performance indicators. Placement statistics are often considered when students and their guardians choose an institution, and there is always stiff competition among institutions for tie-ups with leading firms. In such an atmosphere, the capability of predicting future placement performance and taking proactive measures would spell success for the student and failure otherwise. This is what our system tries to achieve, utilizing placement statistics for future predictive analysis.

Thus, introducing machine learning technologies to alumni management systems makes it possible to change from basic data handling operations to advanced data analysis and processing techniques.

1.2 PROJECT DESCRIPTION

This study explores the design and implementation of an Intelligent Alumni Management System with Machine Learning-based Placement Prediction. The main aim of creating the presented software solution is to develop an advanced platform that would enhance the functionality of traditional alumni management systems by incorporating analytical features.

The system is implemented as a web-based application. To implement it, Spring Boot is used for backend programming, while JSP is utilized to implement the presentation layer. Additionally, MySQL is employed as the relational database management system. The proposed system implements several basic functions such as alumni registration, profile editing, job posting, events arrangement, and collecting feedback. Moreover, the platform allows users with diverse roles to log in. Administrators, staff, and alumni users perform different actions depending on their roles and permissions to ensure secure usage and efficient data organization.

One of the important features of the proposed platform is the inclusion of a machine learning-based placement prediction algorithm that makes it more intelligent. It analyses data related to students' characteristics and attributes, such as academic achievements, technical skills, internship history, project experience, communication skills, etc., to determine their probability of getting placed successfully [6], [7].

The design is scalable and maintainable. By designing the software to have the four components (data management, business logic, user interface, and machine learning) independent, the components can be upgraded or changed independently without having an impact on other parts of the program. That way, even if the machine learning model improves or there is a need for new functionality, it does not mean that the whole platform needs to be changed.

In order to obtain more accurate predictions, several supervised learning algorithms have been implemented, among which there were Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM). Their effectiveness has been compared based on accuracy, precision, recall, and F1-scores. However, despite having high prediction accuracy, some of these approaches showed relatively low F1-score [1], [3], [4]. Therefore, the most effective method was chosen to be applied within the system – the Random Forest approach.

The results produced by this algorithm will help system administrators organize effective actions in order to help students become more employable. They could arrange different training courses, mentorships, etc. in order to make sure that students are fully prepared for job interviews and placement.

In conclusion, the developed Intelligent Alumni Management System significantly outperforms ordinary alumni management platforms by implementing additional analytical solutions. It becomes an indispensable tool for managing alumni data and improving placements.

Alumni Management Systems in Indian engineering colleges haven't undergone much change in terms of features and usability in the last one decade. They merely act as electronic record keepers, allowing for recording graduate data, making announcements related to events and communication purposes. These functions are important but don't suffice the needs of contemporary institutions and do nothing for the benefit of students in pursuing their careers.

The crux of the matter lies in the gap between the large amount of data available with institutions – academic records of the student, certificates and internships completed, project works and placements done previously – and the actionable data derived from them. Placement officers end up making decisions based on instinct and personal experiences, rather than on data analysis. The students who would face problems with getting placement opportunities in the future are not even recognized by these systems.

The key goals of the proposed project include:

1. Development of a web-based Alumni Management System capable of covering all stages of the alumni lifecycle, starting from registration, career tracking, and alumni engagement.
2. Incorporation of the placement probability prediction module based on machine learning techniques capable of analyzing the students' personal data and producing the respective probabilities.
3. Testing of several supervised learning classifiers — Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine — to determine which one performs better for placement prediction purposes.
4. Creation of an administrator interface for presenting the results of the prediction module and allowing for intervention on time.
5. Implementation of role-based access control ensuring proper access to information for the administrators, staff, and alumni.
6. Performance evaluation using traditional performance measures such as accuracy, precision, recall, F1 score, and ROC-AUC to ensure the efficiency of the proposed solution.

7. Provision of a general architecture that may serve as a basis for implementation by other educational institutions interested in predictive analytics in their information systems.

The scope of the project involves developing, implementing, and testing an Intelligent Alumni Management System that will meet the requirements of engineering institutes of Indian Technical Universities. The system will deal with handling alumni information, job listing, events management, and placement information. The scope of the machine learning algorithm includes the classification of students into two classes: 'placed' and 'not placed.'

Data scraping from external sources such as job portals and social networking sites is excluded from the scope of the project but will be implemented later as an improvement. The Intelligent Alumni Management System will be developed as an independent web-based application which can later be migrated to cloud servers. The algorithm will be trained on the past records of placements in particular educational organizations. Thus, generalization to the cases of other universities may require training on another dataset.

CHAPTER 2

LITERATURE REVIEW

2.1 Existing Systems

The traditional AMS systems are mainly intended for storing alumni data as well as coordinating communications and events at educational establishments. In most cases, such platforms are based on the centralized database architecture, and have such basic functions as alumni registration, profile management, records management, and informing about the latest news of the organization. The use of such software allows maintaining long-term relationships with alumni, promoting networking, and organizing various meetings or job opportunities, thus establishing a more robust connection between the alumni and their alma mater.

Although such systems provide basic functionality, they cannot be used for any kind of data analysis or mining. To put it simply, a traditional AMS platform cannot find any patterns or trends, or determine the relationship between alumni or students. It is also important to note that conventional software does not allow performing any predictive analysis, thus limiting the possibility to predict outcomes like alumni employment, job placements, or skill development.

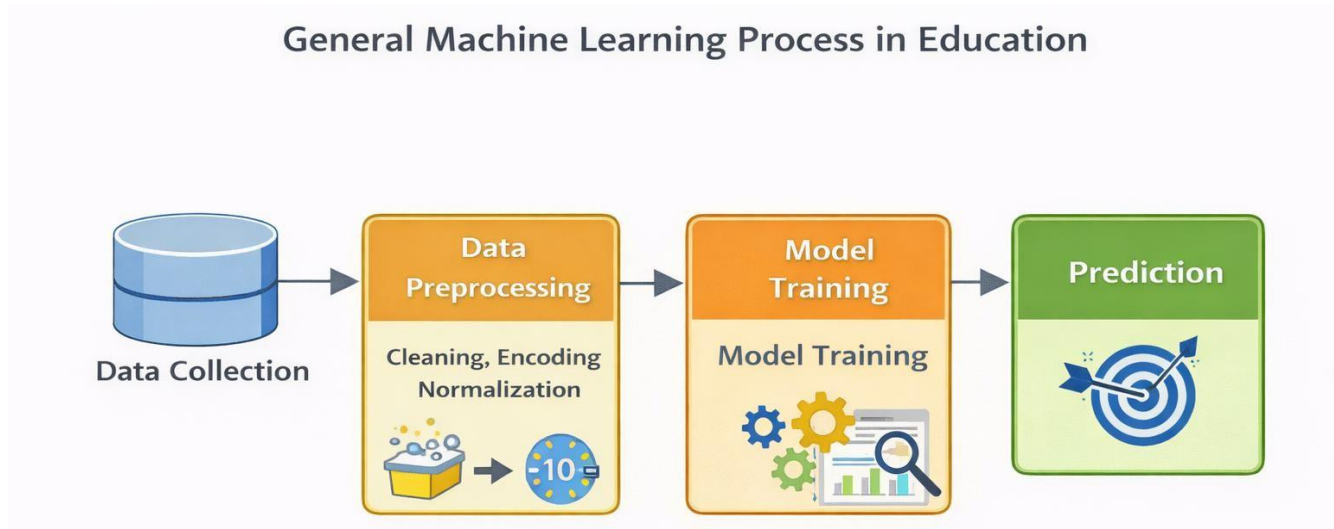
In this context, it should be noted that educational establishments usually conduct all kinds of data analysis manually. As a result, the whole process becomes rather time-consuming, inaccurate, and inefficient. At the same time, the lack of proper tools limits the possibility of using collected information for making decisions and forecasting results.

For instance, certain Indian institute-specific systems that were designed for IITs and NITs have been known to include some features of placement portals in which firms can register themselves, schedule interviews, and keep track of offers made. But again, these are transaction systems and not analytical systems; they only capture what has already occurred without predicting anything about what will occur or recommending any course of action to be taken. This gap between the current transaction systems and decision support enabled by machine learning is exactly what this research aims at addressing.

Taking into account the increasing role of data-driven approaches in the modern education sector, it is obvious that traditional AMS platforms fail to address current requirements. For this reason, there is a need to upgrade such software and integrate advanced technologies like machine learning or data analytics [6], [7].

2.2 Machine Learning in Education

In addition, Machine Learning (ML) becomes increasingly popular for educational data mining as a means of processing student-related data and making predictions concerning both academic success and future placement of students [6], [7]. ML methods allow processing big amounts of data regarding student performance and finding connections that remain undiscovered by using conventional data analysis approaches. As the amount of available digital records that contain information on various aspects of student activity grows, ML can be considered as an adequate tool to obtain valuable conclusions [6], [7].



[Fig. 2.1: General Machine Learning Process in Education]

The above figure shows the standard workflow of machine learning application in educational data analysis. It includes stages of data collection, preprocessing, training, evaluation and predictions, where each stage requires specific efforts from specialists who conduct data analysis. It is important to note that among all listed above stages, data preprocessing is one of the most important steps because proper data preprocessing improves model efficiency and contributes significantly to obtaining accurate prediction results [10].

In fact, machine learning applications within education have been practiced since more than two decades ago when studies were conducted concerning intelligent tutoring systems. However, in recent years, there have been advancements in the practice of applying machine learning because of the existence of extensive data from the educational sector and better computer capabilities. Machine learning techniques have been utilized extensively in different areas related to education, including prediction of dropout cases (Dekker et al., 2009), determination of academically endangered students, exam score predictions, learning resource recommendation, plagiarism detection, and graduate employment prediction.

Machine learning as a process in education follows an established pattern which includes collecting data from institutional databases (learning management systems, student information systems, placement portals), preprocessing data for handling missing data and inconsistencies, training models on collected data, and interpreting model outputs for decision-making within institutions. Each phase of the process comes with its own problems and possibilities.

The supervised learning models that may be used for classification include several types such as Decision Tree, Logistic Regression, Random Forest and SVM. All listed algorithms use several input parameters (student performance, skills, internship, projects, communicative skills) for estimating probabilities of students' placements. Moreover, there are specific strengths of every algorithm, which contribute to making choice between algorithms in case of choosing models for further training. For instance, while Decision Trees generate clear rules in case of classification and Logistic Regression requires little time for calculations, SVM demonstrates high classification accuracy. Random Forest can be considered as an ensemble of several decision trees and helps in achieving accurate results when applying high-dimensional and complex data sets [1], [4].

Model effectiveness may be assessed by using performance metrics such as accuracy, precision, recall, and F1 score. The aforementioned evaluation metrics may help in comparing different models and choosing a proper one for prediction tasks. However, the usage of ensemble methods in some cases demonstrates advantages in comparison with separate models, which can be explained by increased robustness in case of working with high-dimensional and complex data sets [3].

Quite a number of studies in recent times have focused on the particular problem of placement prediction on campus. Several researchers in their studies have tried out data sets featuring variables like secondary school performance, undergraduate GPA, area of study, work experience, and soft skills. From all of these studies, it can be observed that algorithms such as Random Forest and Gradient Boosting, as ensembles, tend to do better than single classifiers if the data set contains both categorical and numeric attributes.

Despite being one of the oldest methods for classification, Logistic Regression has proven itself to have surprising longevity as a model for placement prediction problems. This algorithm's ability to provide interpretable models makes it particularly useful in institutional environments, where placement officers need to know not only the outcome of a prediction but the reasoning behind it. Since the coefficients of a logistic regression model represent the log odds of a feature's contribution to a result, it becomes easy to convey to a student which factors of their profile help or hinder their chances.

Decision Trees provide another level of explainability by creating a set of rules for making decisions that can easily be understood and communicated. In the case of predicting placements, a decision tree will generate rules like 'Students with CGPA above 7.5 having done at least one internship are predicted to get a placement.' While Decision Trees are not necessarily the best model based on overall performance, they prove extremely useful in institutional communication due to their transparency. Unpruned decision trees tend to suffer from overfitting, which is a known weakness of this technique, but ensemble learning addresses this problem effectively.

The Support Vector Machine (SVM) model is very good at performing tasks of placement predictions, especially when there is a small amount of data available and many more features than samples in the dataset. Thanks to the margin maximization idea on which classification by the SVM model relies, there is lower probability of overfitting on the training dataset compared to a tree model. Moreover, thanks to the application of the kernel trick, SVM can identify non-linear connections between feature values and placement. The biggest problem connected to using SVM is connected to probability calibration because raw output values of SVM cannot be interpreted as probability scores.

The Random Forest algorithm, as a collection of independent decision trees that generate output through majority voting, offers both the interpretability advantage of decision trees along with increased generalization capability. In the bagging procedure used by the Random Forest algorithm, the variance is reduced without any increase in bias, and the use of random attribute selection helps decorrelate the trees in the forest, providing additional stability. As for predicting placement, given the nature of input variables such as academic records, skills, and internships, which have potential interactions among themselves, Random Forest appears to be highly appropriate.

In conclusion, ML application in education allows using valuable tools for data analysis and making predictions that help in developing relevant strategies for improving student performance.

EDM is defined as a discipline that makes use of computational techniques for examining educational data, with the objective of discovering useful knowledge to improve educational processes. According to the review made by Romero and Ventura on the topic of educational data mining, the following can be considered as the main tasks in EDM: prediction (predicting some particular target variable), clustering (clustering of students based on their similarities), association analysis (finding relationships among the various variables), student modeling (cognitive and behavioral states modeling), and knowledge extraction for teachers.

The problem of placement prediction can be naturally classified under the category of prediction problems within the EDM domain, but it is additionally intended to serve the purpose of facilitating decision-making in institutions. In terms of the feature sets

used in EDM placement research, their intersection with the features used in this project is fairly extensive and involves: academic performance (such as grades and CGPA scores, as well as backlogs); extra-curricular activities (like participating in projects, hackathons, and certification processes); and practical experience (e.g., internships and workshops).

One factor that needs to be taken into account when analyzing EDM placement data is class imbalance. In cases where the institution under study has a very high rate of students being placed, there will be more 'placed' classes than 'not placed' classes in the dataset, and vice versa. Such an imbalance might cause the algorithm to have high accuracy just because it predicts the dominant class for all observations. Strategies such as synthetic minority oversampling technique (SMOTE), weight adjustment for classes, and stratified cross-validation are usually used to deal with class imbalance. In this study, class imbalance was checked on the dataset, and the data was preprocessed accordingly.

2.2.1 Research Gap

Despite the widespread research on the topic, the majority of papers are concentrated solely on the issue of comparing various algorithms and evaluating them in terms of performance (accuracy, precision, etc.). However, such studies do not delve deeply into the process of implementing predictive models in real-world systems. Thus, a considerable difference can be traced between theoretical research and its practical application, as predictive models are not currently implemented in real-time platforms [1], [4].

According to the literature review, existing Alumni Management Systems do not feature advanced functionality and do not perform analytical operations. These systems allow users to store data, communicate with each other, and manage events, but they cannot provide any recommendations or insights concerning future student success, possible skill gaps, or trends in student performance. Therefore, the institution does not benefit from using data as a resource, as it lacks analytical capabilities [6], [7].

Finally, the field currently lacks an integrated platform for managing alumni lifecycles while employing machine learning-based placement prediction models. Due to the lack of integration, institutions cannot monitor students' readiness, detect potential problems, and correct their mistakes on time. In addition, current systems require additional software and tools for performing analytical operations and making decisions, which may cause inefficiency and poor results.

The study directly addresses the above gaps by developing an end-to-end web application that integrates alumni lifecycle management and placement predictions

using machine learning techniques in real time. Unlike previous studies that only test the algorithms separately, this study incorporates the algorithm with the highest accuracy into a real-world institutional platform and highlights its practicality through screenshots, performance assessments, and workflows.

Thus, there is a demand for an integrated system capable of managing alumni and employing machine learning-based predictions. Such a solution will help bridge the gap between traditional methods of storing and distributing information and innovative approaches to predicting students' success.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 System Overview

The suggested solution will include implementing an intelligent alumni management system with placement prediction, where advanced approaches in data management will be merged with innovative techniques aimed at generating valuable information based on existing and historical data. As a result, it will be possible to make improvements in the placement process as it will take into account relevant historical and current information concerning students. Thus, the platform will provide effective management of alumni, job offers, events, as well as student-related information.

In order to develop the application, it was decided to use a stack of technologies including Spring Boot (for backend), JSP (for frontend), and MySQL (for database). These technologies ensure the high performance and efficiency of the system while interacting between its components and providing necessary functionalities. Besides, the system will allow users to have access to the platform depending on their roles, which may be either administrator, employee, or alumni user. Role-based access will be used to determine the ability of certain users to modify and access particular functionalities. User requests are processed via presentation layer, application layer, and are linked to the data layer in the end.

The implementation of the system is based on the principles of modular design. Every module performs particular tasks that can be associated with layers. Thus, there are the presentation layer responsible for managing the user interface, the application layer that handles logic and requests processing, as well as the data layer containing structured information used in the system operations. Apart from these layers, there is a machine learning layer responsible for performing prediction. Such structure ensures the scalability of the system and makes it more efficient, as illustrated in the architecture diagram.

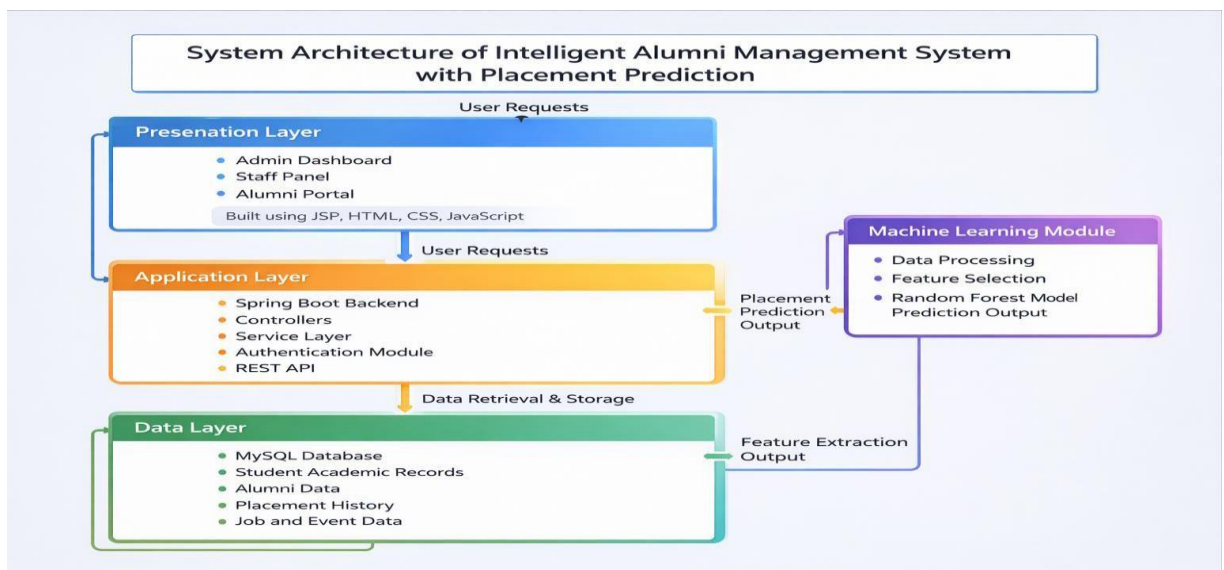
The decision to go for Spring Boot for the backend architecture was driven by a number of technical reasons. The automatic configurations in Spring Boot and the presence of default options minimize the amount of boilerplate code that has to be written, thus enhancing development speed. In addition, since Spring Boot supports REST API development, provides security configurations using Spring Security, and integrates easily with relational databases using Spring Data JPA, it became easy to choose Spring Boot as the framework for the application.

It was decided to use JSP in the frontend implementation due to its strong connection to the Java environment and the possibility of dynamic generation of HTML pages on the server side. Although there are many popular JavaScript libraries that can be used on the client side, such as React or Angular, the advantages of using JSP include the relative simplicity of development and the reduced need for managing states in the application.

Apart from standard functions, there is also a machine learning module that evaluates several student-related parameters like academic performance, technical skills, internships, and project works. On the basis of these parameters, the system uses supervised learning to generate some predictions concerning the probability of student placement.

The information generated as a result is returned to the application layer and represented to the user in the form of a visualized dashboard containing organized and valuable data. Thus, it becomes easier for the administration to understand how ready students are for their placement and how many students will probably require additional training and guidance.

3.2 System Architecture



[Fig. 3.1: System Architecture Diagram]

The proposed system architecture shows the comprehensive structure of the Intelligent Alumni Management System and the manner in which different parts of the system are connected in order to deliver functionalities both in terms of data processing and predictions. The system architecture shows the connections between the different layers of the system, such as the presentation layer, application layer, data layer, and machine learning layer.

In order to design the system effectively, the layered architecture approach is used.

Each of the layers in this type of design performs the certain functions, but the interaction takes place between the layers that are situated adjacent to one another. This approach helps improve the performance of the whole system.

- **Presentation Layer (JSP Interface):**

The presented layer constitutes the user interface of the system. It is built based on the use of the technologies including JSP, HTML, CSS, and JavaScript. The presentation layer provides users with functionalities such as logging in and out, registering, managing the user profile, searching for jobs and predictions, and others. Interaction is made convenient for the administrator, staff member, and alumni user interfaces.

- **Application Layer (Spring Boot Logic):**

The application layer of the system contains the core functionality of the system since it is responsible for request processing, data validation, user session management, and database and machine learning module communication. The functionality provided by the layer is crucial for the operation of the system because of its ability to handle users' data, post jobs, organize events, make predictions, and other tasks.

- **Data Layer (MySQL Database):**

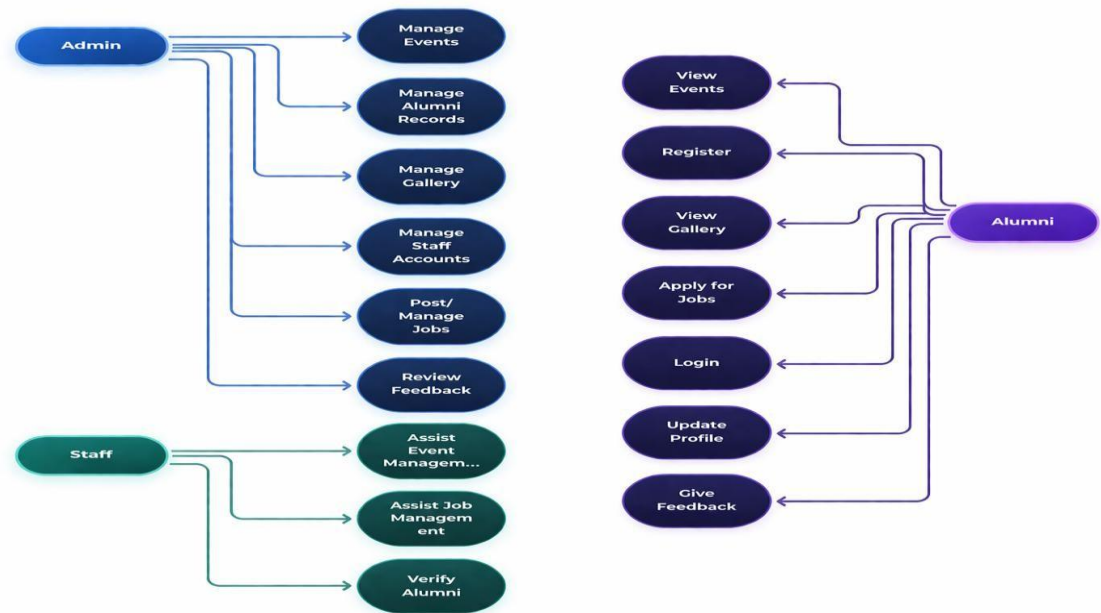
Data layer includes all the structured data required for the operation of the system. In order to perform the mentioned task, the database technology is utilized. In particular, the presented data layer uses the MySQL database management system. Data stored include alumni profile data, students' academic data, skills, internships, job posts, and placements.

- **Machine Learning Layer (Prediction Model):**

The machine learning layer is specifically designed to analyse student data and make placement predictions. Data analysis involves the application of pre-trained supervised learning models such as Random Forest, Logistic Regression, Decision Tree, and Support Vector Machine (SVM) in predicting placement outcomes [1], [4]. The output from the process is transferred back to the application layer for further manipulation and visualization.

The communication between layers is based on an agreed upon data contract. The presentation layer sends HTTP requests to the RESTful API endpoints provided by the application layer. Application layer verifies the requests, processes business logic, and interacts with the database layer to query data, or else it calls the machine learning module via some interface. The machine learning module has been developed as a serialized version of a python-based model which can be called via subprocess/interprocess communication from the main program, allowing for a smooth integration of the two technologies.¹⁴

The entire process flow in all the four layers is well-coordinated in the sense that user inputs are collected using the presentation layer, processed in the application layer, stored/retrieved in the data layer, and finally analysed using the machine learning layer for prediction.



[Fig. 3.2: Use Case Diagram]

The use case diagram shows how various actors interact with the system. Some of the actors include administrators, staff members, and alumni users. Functionalities that can be performed by each actor include registration, login, profile creation/maintenance, job interaction, event interaction, and prediction. Administrators control the functioning of the system through alumni data management, job posting, and prediction result access. On the other hand, staff members support the work of administrators, whereas alumni users interact with the system mainly for profile management and job searching purposes.

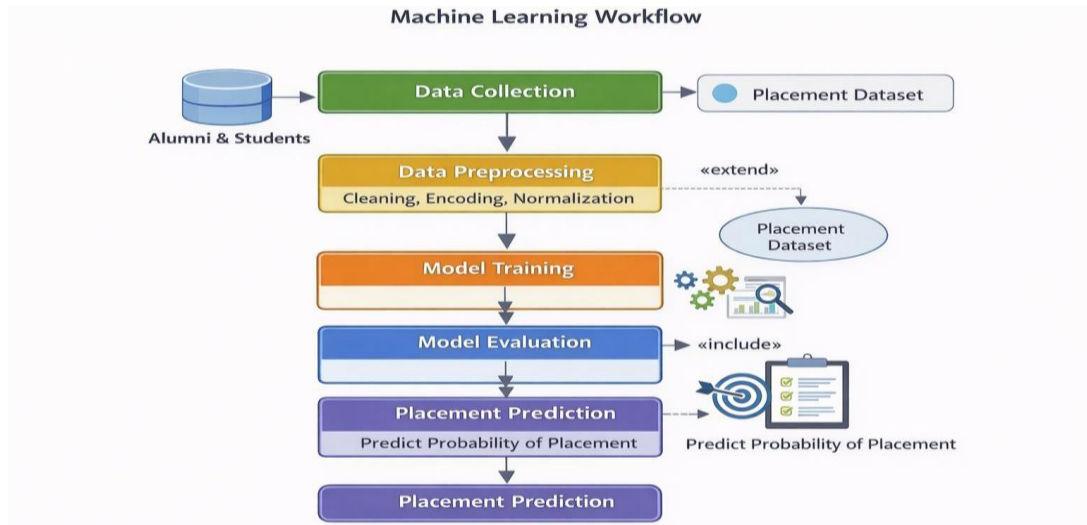
In conclusion, the system architecture showcases how several elements are put together to create a coherent intelligent system.

3.3 Machine Learning-Based Prediction

The placement prediction mechanism that uses machine learning techniques greatly improves the efficiency of the suggested solution. During the placement prediction, numerous student's characteristics, such as grades, technological competence, internship, projects, and others, can be considered the inputs to make relevant predictions. All these characteristics together describe an employability profile of a particular student and help to detect patterns concerning placement results.

In order to guarantee precise and valid predictions, the dataset is required to go

through multiple preprocessing stages prior to training models. Preprocessing includes addressing any problems related to inconsistency and incompleteness, removing duplicates from the dataset, performing data encoding, and normalization. Preprocessing helps to minimize risks of overfitting and improve performance of the models [10]. Preprocessing process can be seen in machine learning workflow.



[Fig. 3.3: Machine Learning Workflow]

Feature Engineering should be considered as an important component contributing towards good performance. The student dataset consists of numerical variables such as CGPA, percent, and no. of projects, categorical variables such as internship (yes/no) and hackathon participation (yes/no), and ordinal variables such as skill rating and communication skill rating. Standardization is performed on numerical features using the min-max normalization technique, whereby all features have to be brought down to the same range to avoid variables of higher magnitude dictating the learning process. Binary categorical variables are transformed to either 0s or 1s while ordinal variables are considered numerical.

The process of feature selection was done to select those features which were informative and remove the redundant and noisy features. Feature selection was done using both the correlation of the features with the output variable and their feature importance obtained from the Random Forest model. Those features which were not correlated with the placement result and had a low feature importance value were deleted. The selected features include CGPA, Technical skills, Communication skills, Internship experience, Number of major projects, Number of mini-projects, Workshop certification count, Hackathon experience, Percentage scored in 10th grade, 12th-grade percentage score, and Number of backlogs.

The use of these algorithms in training predictive models includes applying Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) methods. All these algorithms have unique features; logistic regression is simple and understandable linearly, decision tree is rule-based classification, SVM operates very well in higher dimensional feature spaces, and random forest algorithm uses an ensemble method to increase prediction accuracy [1], [4]. After creating a series of

these models, we conduct experiments to evaluate each of them and choose the most efficient in predicting placement results.

The dataset was divided into training (80%) and testing (20%) portions by stratified sampling such that both portions would have an identical class distribution to that of the whole dataset. The use of stratified sampling is especially critical for placement data sets because of the potential for unbalanced classes in the data set. In addition, five-fold cross-validation was carried out on the training portion.

Model evaluation can be done based on commonly used metrics such as accuracy, precision, and recall scores. The former one represents the total accuracy of our algorithm. The latter two give a more detailed picture of its classification efficiency. These measurements help to evaluate models objectively and choose the most effective among the tested algorithms.

As a result of conducting experiments, Random Forest algorithm proved to work best in prediction. An ensemble method that combines many decision trees significantly reduces the risk of overfitting and improves model generalization. In addition, the model is highly capable of dealing with non-linear problems [3]. Due to its performance, Random Forest algorithm is chosen as the final one to be integrated into our system.

For Random Forest being chosen as the final deployment model, the process of hyperparameter optimization through the technique of grid search with cross validation was executed. The main hyperparameters that were tuned included the number of trees (`n_estimators`: 50, 100, 200), tree depth (`max_depth`: None, 10, 20, 30), minimum number of samples needed to split a tree node (`min_samples_split`: 2, 5, 10), and the number of features used for splitting (`max_features`: 'sqrt', 'log2'). The best combination of these hyperparameters turned out to be `n_estimators=100`, `max_depth=None` (grown to full depth), `min_samples_split=2`, `max_features='sqrt'`.

The implementation of a prediction algorithm gives the ability to process student's data dynamically and get valuable insights from it. Thus, we can determine which candidates are likely to be placed successfully and which need further preparation for being considered.

3.4 System Workflow

User data → Database → ML Model → Prediction → Dashboard

The system workflow encompasses all the procedures involved, from accepting the initial user input data to generating predictions. In the first place, the input data includes the information related to students, which can be characterized by academic achievements, technical proficiency, internship experience, and other pertinent variables. Such data is received and accepted via the interface and is stored within the database. The effective management and storing of data are an important prerequisite to performing further analysis.

Further on, the stored data is transmitted to the machine learning module, where preprocessing and feature extraction takes place as part of preparation of the dataset [10]. Following such preparation of the data set, it will be used by machine learning algorithms that will analyse its features to produce prediction results [1], [4].

As the output from the above-mentioned procedure, the system generates predictions, which are subsequently delivered to the application layer and represented in the form of visualization on the dashboard. It is important to note that due to such a presentation, it is easier for the administrator and staff members to understand the results of analysis.

In general, one may state that the workflow involves effective data management alongside machine learning-based analysis. Through this process, institutions are able to effectively assess the data related to students and use data-driven decisions for improving their placement [6], [7].

CHAPTER 4

RESULTS AND DISCUSSION

The suggested Intelligent Alumni Management System was successfully developed and tested to evaluate its efficiency in terms of functionality and efficacy of the machine learning-based integrated placement prediction module.

The app performs important functions related to managing information about alumni users, including registration of alumni users, management of alumni profiles, management of job postings, scheduling of events, and management of alumni feedback. The platform includes a mechanism of role-based security control, which means that the administrators, staff members, and alumni users have access to the system based on predefined access permissions and can use it for its proper functioning. The interactive UI is provided for better user experience.

For placing students according to their professional skills and preferences, several machine learning models (Logistic Regression, Decision Tree, SVM, and Random Forest) were implemented and tested. The models were trained using a pre-processed student dataset, and performance criteria for all models were calculated for further comparison of results [1], [4].

Performance comparisons are provided below.

Algorithm	Accuracy (%)	Precision	Recall	F1-Score
Logistic Regression	93.1	0.91	0.93	0.92
Decision Tree	93.0	0.92	0.91	0.91
Support Vector Machine (SVM)	93.6	0.92	0.93	0.92
Random Forest	94.2	0.93	0.93	0.93

[Table 4.1: Performance Comparison of Machine Learning Models]

As is evident from the findings, Random Forest model outperforms others on all evaluation criteria. It demonstrates higher accuracy and shows consistent success in performing classification based on complex dataset.

As can be seen from the results, Random Forest demonstrated high accuracy (94.2%) together with good precision and recall. Despite relatively good performance of

Logistic Regression, Decision Tree, and SVM, Random Forest was much more effective in classification of students into categories "placed" and "not placed". The superiority of the model lies in its ensemble technique that uses multiple decision trees in prediction [3].

It should be mentioned that, in general, ensemble methods demonstrate higher stability in prediction than single-model methods. Using the outputs of multiple trees, Random Forest ensures lower probability of overfitting and higher generalization in case of new datasets [3].

Another significant finding resulting from the comparative analysis of the models' performance is the fairly low difference in their results. All four models manage to reach more than 93% accuracy, indicating that the features used in this analysis are relevant enough when it comes to predicting placement outcomes (CGPA, skills, internship experience, participation in projects, and others). Furthermore, such similarity in performance can be explained by the fact that the input data pre-processing phase allowed for developing a quality input that helped all classifiers perform well despite the differences in complexity. In turn, it should be emphasized that Random Forest outperforms the second best classifier (SVM) by 1.2 percentage points, which may not seem much in numbers; however, it means 12 extra students classified in case of 1,000 students.

The ROC-AUC measures offer a more detailed insight into the performance of the models compared to using the accuracy score alone. For instance, the Random Forest classifier's AUC of 0.9868 is an indication that the model can achieve perfect discrimination of placed and not-placed students at any decision threshold. This is especially significant since, in practical applications, the threshold used in determining whether a student should be considered at risk will vary according to organizational needs. An organization that seeks to ensure no at-risk students go undetected (high recall) can use a lower threshold, while those focused on precision can opt for a higher one.

The process of prediction is based on specific variables that should be considered important input features.

Features	Description
CGPA	Score related to academic achievement
Skills	Skills based on technical ability
Internship	Experience gained from internships
Projects	Number of projects undertaken
Communication	Skills related to communication
Placement Status	Target feature (Placed/Not Placed)

[Table 4.2: Sample Student Dataset Features]

These factors greatly affect the placement process and are important for increasing the precision of the model. The analysis shows that factors like academic achievement, technical abilities, internships, and projects have a huge impact on student placement, similar to what has been stated in educational data mining literature [6], [7].

The feature importance analysis done on the trained Random Forest model showed that the CGPA score was the most influential feature in predicting placement success, explaining about 23% of the total information gain in the model. This is in agreement with the general notion that academic performance is one of the major factors that recruiters use as a screening tool. The technical skills rating and communication skills rating were found to be the next two most important features, together explaining another 30% of the information gain. The internship experience was a binary variable, yet very influential in determining whether a student would be placed successfully. Students who had internship experience had better placement rates compared to students without internship experience, regardless of their academic performance.

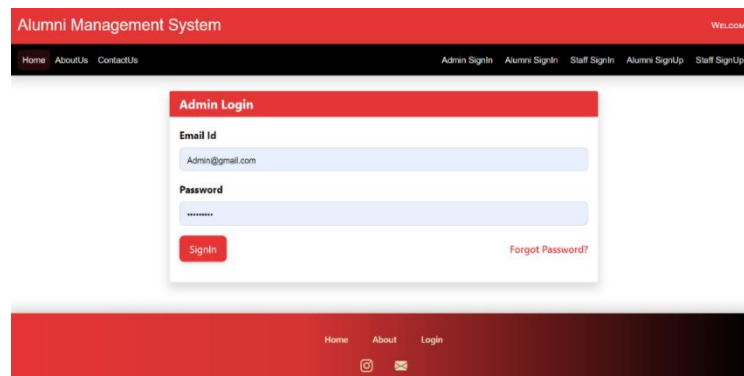
Using the prediction tool in the framework allows organizations to easily distinguish between students with good placement prospects and students who need further training.

In conclusion, the system proposed by us improves the existing methods of alumni management through the implementation of predictive analytics tools.

4.1 System Interface and Screenshots

The designed system provides a user interface that is easy to use by various users like administrators, staff, and alumni. The following pictures highlight the various features of the system.

1. Login Page

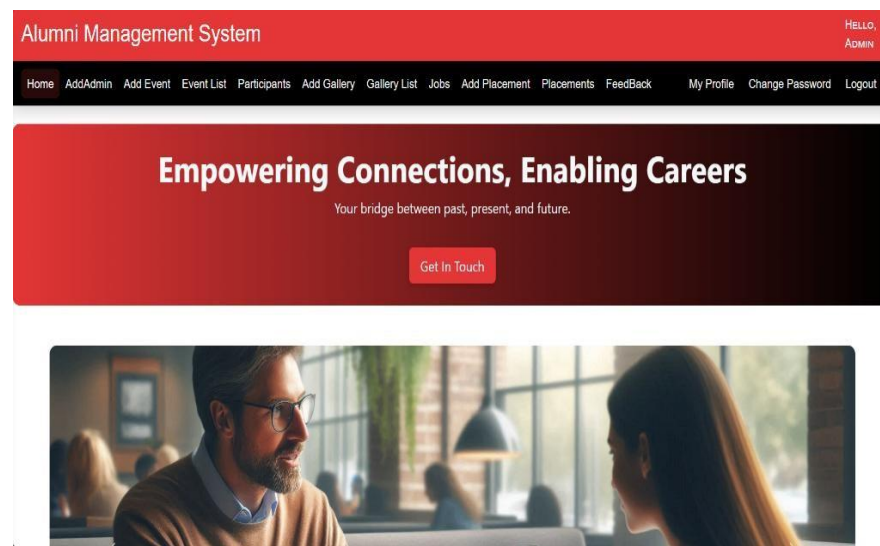


[Fig. 4.1: Login Page of the System]

The log-in screen is used to ensure that users access the system through their usernames and passwords. This will make sure that users are authenticated and directed to various screens based on their role whether it be admin, staff, or alumni.

The login screen is kept simple to maintain a neat and professional look that matches the institution's branding. The 'Forgot Password' link can be used to reset passwords by verifying emails. The application incorporates the security measure of preventing excessive login attempts; if there are five failed login attempts in succession, then the user is automatically locked out for 15 minutes. Navigation menus based on roles are automatically displayed depending on which user role logged into the system.

2. Dashboard



[Fig. 4.2: Admin Dashboard]

The admin dashboard gives the status of what is happening in the system, for instance, the statistics of the alumni database, jobs that have been posted, and events in the system. In addition, it makes it possible for admins to manage various things in the system including accessing the prediction results [6].

The purpose of the dashboard is to enable situational awareness at a glance for both the placement coordinator and the institution administrator. Performance metrics that will be shown on the dashboard include number of registered alumni, number of active job openings, future events within the next 30 days, and a placement readiness report for the current batch of graduates. The placement readiness report will be generated using the results of the predictive model applied on all current student profiles, showing a summary of how many students belong to the high probability group ($> 70\%$), medium probability group ($40\text{-}70\%$), and at risk group ($< 40\%$).

3. Profile Page

The screenshot shows the 'My Profile' section of the Alumni Management System. At the top, there is a red header with the system name and a user greeting. Below this is a black navigation bar with various menu items. The main content area has a red header for 'My Profile' and a profile picture placeholder. Below the picture are input fields for Name, Email, and Contact No, each with a corresponding label. At the bottom of the form are 'Save' and 'Reset' buttons.

Field	Value
Name	Admin
Email	Admin@gmail.com
Contact No	96365XXXXXX

[Fig. 4.3: Admin Profile Management]

This part is used by admin users to manage their information. Personal, academic, and professional information can be added and managed.

The profile management tool enables students to have elaborate profiles with regard to their capabilities, which can include everything necessary for predicting placements. Students' academic profiles, such as semester grades and backlogs, can be entered as data files or input manually. Endorsements of skill levels by instructors or corporate mentors can also be included by appropriate individuals who can verify the claims made by the students themselves. The tool also includes an extracurricular activity timeline, where students can list any certificates or internships they have had.

4. Job Page

The screenshot shows the 'Job List' section of the Alumni Management System. It features a red header with the system name and a user greeting. Below this is a black navigation bar with various menu items. The main content area has a red header for 'Job List' and a search bar with 'Search By Title' and 'Search By Location' options. Below the search bar is a table listing jobs with columns for job ID, title, location, and a 'Delete' button. At the bottom right, there are 'Previous' and 'Next' buttons.

Job ID	Job Title	Location	Action
1. tcs	SDE		Delete
2. tcs	backend dev		Delete
3. tcs	social manager		Delete

[Fig. 4.4: Job Management]

Job posting and event management is done by the admin or staff members using this part of the software. Job posts are then accessed by the alumni while events can also be attended by them.

Through the job management section, job placements can be listed in detail with regard to the title, the qualification that is necessary, what qualities are advantageous, how much pay the position would offer, and the deadline for applications. The system also has an option of eligibility filtering on a per job basis, meaning that the system will help determine which of the students satisfy the minimum requirements for the job available. Alumni are also informed regarding job positions, creating the flow between the institution and its alumni.

5. Prediction Output

```
y_pred = pipeline.predict(X_test)

print("First 5 Test Predictions:")
print(le.inverse_transform(y_pred[:5]))

# new student data
new_student = pd.DataFrame([
    'CGPA': 8.5,
    'Major Projects': 2,
    'Workshops/Certifications': 3,
    'Mini Projects': 2,
    'Skills': 7,
    'Communication Skill Rating': 4,
    'Internship': 'Yes',
    'Hackathon': 'Yes',
    '12th Percentage': 85,
    '10th Percentage': 80,
    'backlogs': 0
])

pred = pipeline.predict(new_student)

print("\nNew Student Result:")
print(le.inverse_transform(pred)[0])

First 5 Test Predictions:
['Placed' 'NotPlaced' 'Placed' 'NotPlaced' 'NotPlaced']

New Student Result:
Placed
```

[Fig. 4.5: Placement Prediction Output]

This part contains the output from the model which is the placement probability of students in the system [1], [4].

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The research led to successful implementation of Intelligent Alumni Management System in conjunction with Machine Learning-based Placement Prediction that showed effective usage of data and predictions. The system represents the enhancement of a traditional alumni management system turning it into a more efficient and smarter platform.

The web application is built using Spring Boot, JSP, and MySQL technologies and possesses features necessary for any alumni system such as student registration, management of profiles and events, publishing of job openings and others. Usage of machine learning makes it possible to make accurate predictions about student placement according to academic performance, knowledge of technical skills, internships, and projects. Thus, the system is aimed at changing traditional record-keeping into intelligent analysis of the data [6], [7].

Among the tested algorithms, Random Forest showed the best results providing high accuracy and performance, which means it could be applied effectively as the basis for placement prediction. Being an ensemble method, it provides stable and consistent predictions and reduces the likelihood of overfitting. Moreover, the system manages not only to recognize students with good placement opportunities but also students requiring additional training [3].

From the confusion matrix analysis, it can be seen that the strength of the Random Forest model lies in its ability to reduce the number of false negatives, which refer to instances where students at risk are mistakenly predicted to have a higher likelihood of placement. In the organizational environment, this is the most expensive form of mistake, as it causes the organization to miss out on students who require intervention. This means that the chosen model is very appropriate for the use case scenario.

Integration of predictive analytics into alumni management system makes it possible for institutions to adopt proactive approaches to placement and take steps for the enhancement of student skills and abilities needed for successful employment.

The project also shows how the combination of machine learning models in Python with web applications in Java is possible with the correct inter-process communication protocols. The design pattern used here ensures that the two components use their

strengths while working together as one system. The Spring Boot framework is used to implement enterprise-level web application features, while Spring Security is used for authentication.

Overall, the project demonstrates successful use of machine learning technologies allowing building predictive systems to enhance educational process, make decisions on placement, and further growth of an institution.

5.2 Future Scope

Further improvements could be made by using even bigger datasets obtained from multiple institutions, making the predictions even more accurate and improving the generalization capabilities of machine learning models. Some additional features, such as an analytics dashboard, automatic notifications, or mobile app support, can be implemented to increase usability and accessibility.

It is also possible to implement more advanced machine learning algorithms such as deep learning, ensemble optimization, or other innovative solutions that will help to improve prediction accuracy and performance [1], [3]. The integration of the system with some cloud platform will allow for scaling the solution, increasing accessibility and efficiency.

Also, the system can be enhanced by integrating it with institutional infrastructure such as ERP or LMS systems. This way, it will be possible to exchange data and manage all workflows within these systems efficiently.

Another potential upgrade would be to introduce personalized recommendations and interventions that would help students identify their skills' weaknesses and work on improving them to increase employment chances after graduating from college.

Another possible avenue for improvement is the incorporation of real-time data feeds. At present, students have to input their own details into the system. In future versions, the software could be integrated with a Learning Management System, which would allow for the automatic inclusion of assignment scores and course performance metrics, while online coding tools like HackerRank or LeetCode would provide information about the individual's technical abilities. LinkedIn's API would allow for tracking alumni progress in the workforce.

Finally, the ability of the system to be extended to facilitate multi-institutional cooperation would be a promising long-term objective. With the system's federated version able to gather anonymous data from different institutions, training of models that are more generally applicable can take place without compromising the data's confidentiality at the participating institutions. This kind of collaboration may also allow for the benchmarking of results, where each institution can evaluate their own performance in terms of placements compared to other institutions.

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APPENDIX 1

A. Sample Code Snippet

```
@RestController
@RequestMapping("/api/alumni")
public class AlumniController {

    @Autowired
    private AlumniService alumniService;

    @GetMapping("/all")
    public List<Alumni>
        getAllAlumni() { return
            alumniService.getAllAlumni();
        }

}
```

B. Database Schema

Database used by the system is MySQL. Some of the key tables are:

- Alumni (id, name, email, batch, job information)
- Student (id, marks, skills, internship, placement)
- Jobs (id, company, designation, eligible criteria)

C. Sample Prediction Output

Input:

CGPA: 8.5, Internship: Yes, Skills:

Good Output:

Probability of Placement: 92% (High)

RESEARCH PAPER

An Intelligent Alumni Management System with Machine Learning-Based Placement Prediction and Predictive Analytics

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Abstract—

The Alumni Management Systems (AMS) plays a support system in educational institutions because it enables the institution to engage with the alumni on long-term basis, record graduates, assist institution in networking and career development of alumni. Although classic AMS platforms serve well in registration, communication and event coordination functions, they are mostly centralized data repositories and do not provide intelligent analytical functions. Within the framework of the growing competitiveness in the field of higher education, the growing significance of placement statistics, institutions are in need of predictive and data-driven systems beyond the simple management of records. This study provides a proposal of an Intelligent Alumni Management System with a Machine Learning-based Placement Prediction and Predictive Analytics module. The suggested system is a combination of a web-based alumni management system built with the Spring Boot, JSP and MySQL and a supervised machine learning system to predict student placement results. The predictive module uses past student data in the form of academic performance, technical, internship, projects, and communication skills among other pertinent features to determine a likelihood of placement.

Several classification algorithms are used, i.e., Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) and compared to one another to define the most successful predictive model. Systematic preprocessing of the dataset is done by cleaning the data, encoding categorical variables, feature selection, normalization, and train-test splitting to guarantee the model robustness and reliability. The metrics of performance evaluation deployed to determine the efficiency of the classification include accuracy, precision, recall, F1-score, and confusion matrix analysis, and the predictive analytics implemented in the Alumni Management System allow identifying high-placement potential and low-potential students in time and adopting the appropriate approach to academic or skills-based support. Moreover, the system offers analytical administrator dashboards, which assist in strategic planning, resource allocation, and placement management based on data. As theoretical results show, ensemble-based learning methods are more effective than traditional classifiers, underscoring the usefulness of machine learning in predicting educational outcomes. The proposed framework can convert a traditional alumni management platform into a smart decision-support system, which is more scalable, flexible, and more transparent to the institution. This system provides a holistic, future-proof solution to the contemporary educational institutions, by connecting alumni lifecycle management to predictive placement analytics.

Keywords—Alumni Management System, Machine Learning, Placement Prediction, Predictive analytics, Random Forest, Decision tree, Logistic regression, Support Vector machine, Educational Data Mining, Student Performance Analysis.

I. INTRODUCTION

Over the past few years, the greater educational institutions have paid more attention to the enhancement of alumni interaction and greater placement rates as the primary performance criteria of the academic excellence. The alumni networks are considered to play a significant role in the growth of an institution by means of mentorship, industry partnership, sharing of knowledge and making of a name. At the same time, campus placements are highly important to influence the professional career of students and increase the institutional brand value. The capacity to have organized records of the alumni coupled with the enhanced performance in regard to placement has proved to be a strategically sensitive issue as the institutions compete fiercely.

The conventional Alumni Management Systems (AMS) are basically created to handle the registration of alumni, communications, and organization of events, postings, and maintenance of records. These systems are normally on centralized database architecture which is efficient in terms of storage and retrieval of alumni data. Nevertheless, the majority of the traditional AMS interfaces are management tools that are not intelligent and cannot perform analytical functions. There is no predictive information given about the likelihood of student placement, skill deficiencies or performance patterns. This implies that the decision-making process in institutions does not depend on data forecasting as it is more likely to depend on manual analysis and past experiences.

Due to the fast development of the fields of Machine Learning (ML) and Educational Data Mining (EDM), predictive modeling has become an influential tool to analyze academic data and determine the outcomes of students [6], [7]. Machine learning algorithms are able to determine the unidentified patterns in past academic history and placement data and come up with meaningful predictions [1], [2]. Predictive analytics can help an institution to recruit the student with high placement potential, identify at-risk

students, and suggest specific interventions to enhance employability in the context of the campus recruitment.

The current study suggests Intelligent Alumni Management System with a Placement Prediction module based on the machine learning. The goal is to change an old record-based AMS to a data-based decision-support platform. The system we have is a web-based alumni management system that has been developed with use of the Spring boot, JSP and MySQL with the use of the supervised machine learning classifiers like Logistic Regression, Decision Tree, Random Forest and Support Vector machine [3], [4]. The predictive model is based on the academic performance indicators, technical skills, internship experience, project exposure, and other pertinent qualities to project placement probability.

II. LITERATURE REVIEW

In the recent years, machine learning as an educational data analysis method has been of great interest [6]. There has been interest among researchers in predictive methods of modeling in determining academic performance, student retention, and placement outcomes [7]. Specifically, placement prediction has become a significant field of research because it directly affects institutional reputation and career formation of students.

To start with, some studies have leveraged the supervised machine learning algorithms to forecast student placement status on the basis of academic and demographic characteristics [7], [16]. One of the most popular methods is classification methods in the form of Decision Tree, Naive Bayes, and Random Forest that would classify students as either being placed or not [8], [9]. To provide an example, the previous studies found that the ensemble learning algorithms like Random Forest tend to perform better than the single classifiers because they reduce overfitting and can deal with nonlinear dependencies in the data [3]. Accuracy, precision, recall, F1-score and confusion matrix are commonly used as comparative measurements of the model performance [10].

Algorithms like Support Vector Machine (SVM), Logistic Regression and K-Nearest Neighbor are also other algorithms that have been implemented in other studies to make placement forecasting [4], [1]. These methods are based on structured data sets that include the attribute of the secondary and higher secondary percentages, undergraduate scores, specialization, work experience, and technical competencies. The findings of literature reveal that most of the experiments conducted on ensemble-based classifiers and other hybrid models have high predictive reliability than the single-model prediction methods [3].

Although significant work has been conducted on the areas of placement prediction and educational analytics, a gap in the literature on the research is observed on the construction of an integrated system, which would unite the alumni lifecycle management and real-time placement prediction. Most of the previous studies concentrate only on the algorithmic comparison without integrating the predictive model to a scalable web based institutional platform. Thus, the study fills the gap by suggesting an AI-based Alumni

Management System that combines the management of records, predictive modeling, and analytical visualization into a single architecture.

III. METHODOLOGY

The Intelligent Alumni Management System proposed is meant to loosely couple traditional alumni lifecycle management with a machine learning-based prediction framework of placements to create a single and scalable institution platform. The main focus of the system is to reduce the gap between data management and predictive analytics by introducing intelligent forecasting directly into the alumni ecosystem. The system is designed in such a way that it uses a modular and layered architectural approach that guarantees maintainability, extensiveness and optimal operational performance. Incorporating the web-based application development and supervised machine learning methods [1], [2], the platform converts a classic record-management solution into a proactive decision-support solution.

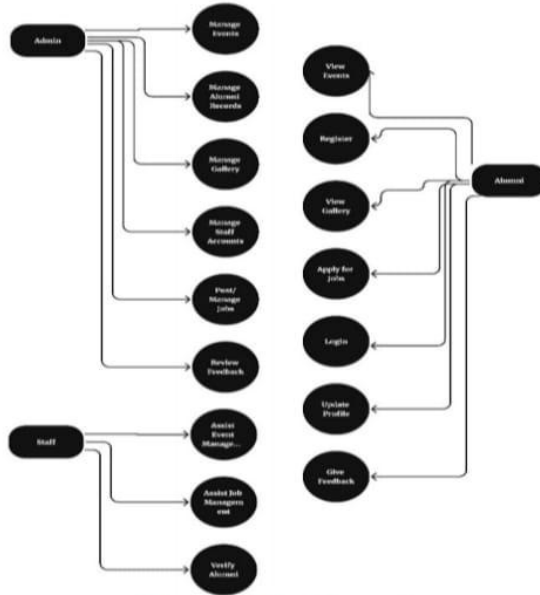
The system is being built on Spring Boot Framework as a backend, JSP as presentation layer and MySQL as the relational database management system. All of these technologies form a powerful platform to handle records of the alumni, employment opportunities, arranging of events and placing the students. The backend does the authentication, business logic, and database communication whereas the front end provides an interactive and easy to use interface to the administrators, the staff members, and the alumni users. The academic data, competency related features, internship history, project experience, and placement history are organized in the structured database, which are essential inputs to the predictive analytics module.

The main innovation of the system is that it integrates a Machine Learning-based Placement Prediction module. This analysis element derives structured scholarly and skill-based variables out of the database and processes them by supervised classification algorithms [1], [2]. The model uses the trend in past placement data to give probability scores that place students in either the high, moderate, or at-risk placement groups. Ensemble methods like the Random Forest are especially useful when dealing with nonlinear complex relationships in such data [3]. The visual dashboards used to display the prediction results help administrators as well as staff members to track the overall placement readiness in batches.

The design requirements of the proposed system are focused on scaling, modularity, and data-driven intelligence. By integrating predictive analytics in the alumni management system, it does not only improve the effectiveness of its operations but also provides a formalized method of predicting the success of the placement. The integration helps institutions to cease reactive placement management and adopt proactive and strategic planning, which will ultimately enhance student employability and institutional performance.

The operational interactions between the administrators, staff members and alumni users are depicted using the Use Case Diagram that is shown in this section. The diagram illustrates

the operations of a role and the window panes of different categories of users have been well defined. The administrator has permission to control events, alumni data, gallery data, personnel, employment opportunities, and response to feedback. The staff position helps with the administrative work through helping in the management of events, job coordination and alumni verification. The system interacts with alumni users by allowing their registration, log-in, profile updating, viewing events and gallery content, job application, and feedback. The graphical representation of the user activity provides a well-defined segregation of duties and role-based access control on the system. More so, the diagram on how the operational workflows interrelate and still maintain modular independence supports the scalability and maintainability of the proposed architecture.



[Fig. 1. Use Case Diagram]

IV. MACHINE LEARNING-BASED PLACEMENT PREDICTION

A. Dataset Description

The dataset to be applied in prediction of placements is structured records of students with academic and skill-based essence. These features are academic performance variables like CGPA and percentages, technical expertise, experience in internship, projects, certifications, and communication skill. The target variable is the placement status of students which is either placed or not placed. All these characteristics can be used to inform the meaningful knowledge about the employability profile of every student and are the input variables of supervised classification models [8], [9].

B. Data Preprocessing

Both the systematic preprocessing and the quality and consistency of the dataset are done prior to the model training. Missing data are addressed accordingly; continuous variables are coded into numeric data and unnecessary

features are dropped in an effort of enhancing efficiency of the model. The feature scaling is used where it is needed to standardize features of the numerical variables. The data is further categorized into training and testing data in order to test the model generalization abilities. This is a preprocessing pipeline that is structured to boost prediction reliability and reduce bias in prediction performance [10].



[Fig. Machine Learning Algorithm Dataflow Diagram]

C. Model Generation and Classification

In order to identify the most efficient predictive model, four supervised machine learning classifiers have been developed and compared: Logistic Regression, Decision Tree, Random Forest, and Support Vector Classifier (SVM) [1], [4]. All the models were trained on the processed data and assessed with regular performance values such as accuracy, precision, recall, F1-score, and ROC-AUC [8], [9].

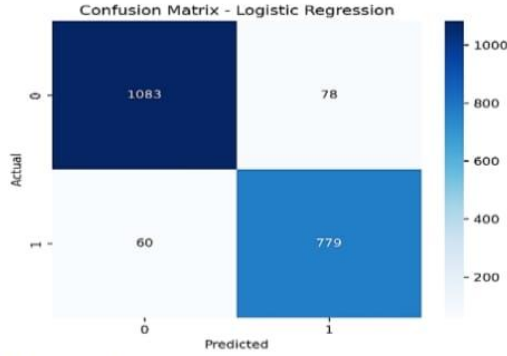
1) Logistic Regression

The Logistic Regression was used as a base linear classification model to predict the likelihood of a student being placed depending on weighted contribution of features [1]. The algorithm estimates the correlation between independent variables and binary placement outcome, with the help of a sigmoid activation function. The model had accuracy of **93.10%**, precision of 0.9089, recall of 0.9284 F1-score of 0.9186 and ROC-AUC of 0.9811. Fig. represents the confusion matrix of the Logistic Regression. The high recall value shows that the model is effective in identifying students who are likely to be placed and also, the precision is balanced Sigmoid function:

$$P(y = 1|x) = \sigma(w^T x + b) = 1/(1 + e^{- (w^T x + b)})$$

Binary Cross Entropy Loss function:

$$L = - (1/n) \sum [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$



2) Decision Tree

Decision Tree classifier is developed to build hierarchical decision rules by subdividing the dataset according to the importance of features [16]. The methodology ensures interpretability through a clear delineation of boundaries of decision-making that is based on academic and skill qualities. The model had an accuracy of **93.00%**, a precision value of 0.9185, a recall value of 0.9141, an F1-score of 0.9163 and a value of ROC-AUC of 0.9278. The Decision Tree classifier confusion matrix is given in Fig. Despite the good performance of the model, there might be some slight overfitting effects since the model can influence its generalization in comparison to ensemble methods.

Entropy:

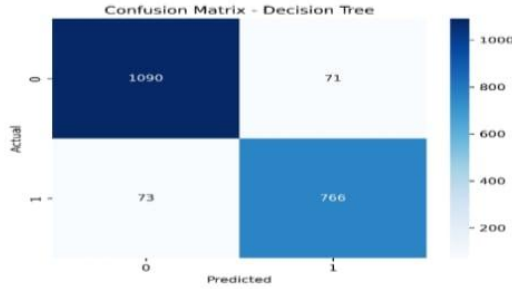
$$H(S) = -\sum p_k \log_2(p_k)$$

Information Gain:

$$IG(S,A) = H(S) - \sum (|S_v|/|S|) H(S_v)$$

Gini:

$$Gini(S) = 1 - \sum p_k^2$$

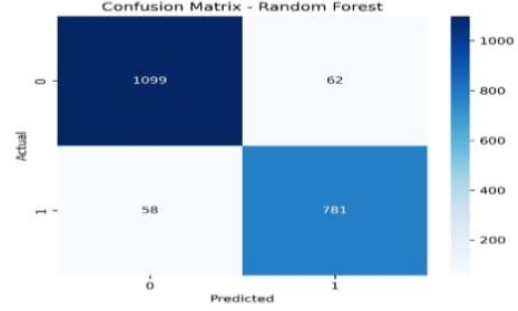


3) Random Forest

Random Forest is an ensemble learning algorithm that takes a combination of decision trees and is used to aggregate with majority voting [3]. This method eliminates variance and enhances robustness of prediction. The highest accuracy was **94.20%**, precision of 0.9298, a recall of 0.9320, F1-score of 0.9309, and a ROC-AUC of 0.9868, which all belong to the Random Forest classifier. Fig. represents the confusion matrix of random forest. The higher ROC-AUC score means that there is a high discriminative ability among non-placed and placed students. The ensemble mechanism is good at capturing the complex nonlinear relationships of the dataset

and therefore the ensemble mechanism attains higher predictive results.

$$\hat{y} = \text{mode} \{h^1(x), h^2(x), \dots, h(x)\}$$



4) Support Vector Machine (SVM)

Support Vector Machine classifier determines the most optimal hyper plane which maximizes the margin between the placement classes in the feature space [4]. This method works well especially with high dimensional data. The SVM model was **93.65%** accurate and had a precision 0.9188, a recall value of 0.9308, F1-score value of 0.9248 and an ROC-AUC value of 0.9831. Fig. represents the confusion matrix of SVM. The high ROC-AUC indicates that it is highly classified with the highest classification being the case but it has a little less accuracy compared to the random forest classifier.

Optimal hyperplane:

$$w^2x + b = 0$$

Margin maximization (hard margin):

$$\min_{w,b} \frac{1}{\|w\|}$$

Equivalently, the primal optimization problem:

$$\min_{w,b} \frac{1}{2} \|w\|^2$$

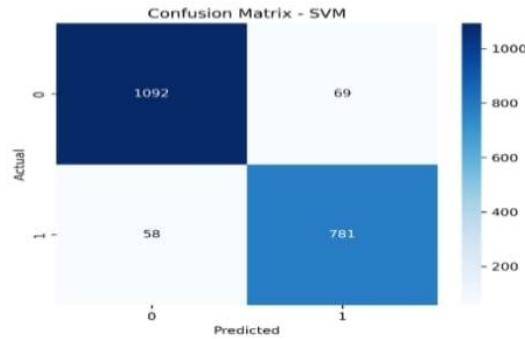
Constraint:

$$y_i(w^T x_i + b) \geq 1$$

Soft Margin:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

Where $\xi_i \geq 0$ are slack variable and C controls the margin-error trade-off.



D. Performance Comparison

According to a comparative analysis of the four classifiers, ensemble-based learning methods are more effective than single-model classifiers in the task of placing objects [3]. Random Forest was the most accurate and the one with the highest ROC-AUC (0.9868) and, as such, it has a higher predictive stability and is more robust than the other evaluated algorithms. Although the results offered by Logistic Regression and Decision Tree are competitive, have high interpretability and the SVM is high in terms of classification margin efficiency, the results offered by the Random Forest are the most appropriate with respect to precision, recall, and overall accuracy.

These findings confirm the power of machine learning in learning outcome prediction [6], [7] and support the fact that the Random Forest classifier should be incorporated into the Intelligent Alumni Management System of the proposed Intelligent Alumni Management System in terms of real-time placement prediction.

V. SYSTEM ARCHITECTURE

The proposed Intelligent Alumni Management System is developed based on a layered and modular system architecture that guarantees scalability, maintainability and smooth integration of predictive analytics. The architecture has four major layers, which include the Presentation Layer, Application Layer, Data Layer and Machine Learning Integration Layer. All layers have separate duties and have well organized communication with the neighboring layers.

Presentation Layer is created on the basis of JSP and such web technologies as HTML, CSS, JavaScript. This level is the user interface of the administrators, staff members, and the users of the alumni. It offers profile management, job application dashboards, event view dashboards and analytical visualization of placement forecasting. The role-based access control systems make sure every category of users only communicates with the functionalities that he or she is supposed to communicate with in the system.

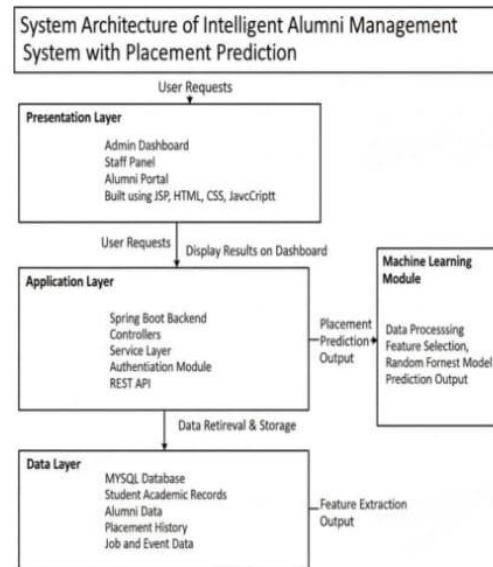
Application Layer is done in Spring Boot framework which is a business logic implementation as well as authentication, session management and API communication. This layer is the one that translates user requests to the backend database and the reverse so that proper service operations are carried out. The entire modules such as the alumni registration, job management, events coordination, and feedback handling are

handled in this layer using organized controller and service classes.

The Data Layer will make use of MySQL as the relational database management system. It bears structured data such as the academic history of the students, their abilities, internship experiences, placement history, and user accounts. Relational mapping is necessary to have data integrity and performance queries. The database is the Centre where the machine learning module retrieves the appropriate features to predict in the modeling.

Specialized Machine Learning Integration Layer is introduced to provide usage of placement forecasting. This layer takes the attributes which are selected and access the database and feeds them into the trained predictive model. The deployed Random Forest classifier goes through the input features and produces a placement probability score [3]. It is then relayed to the application layer and represented on the administrative dashboard in organized fashion. This assimilation will result in the dynamism of prediction generation through the updated student information.

The general flow of data starts with the input of the user in the presentation layer, and then the data is validated and processed in the application layer. The appropriate academic and ability characteristics are read off the database and sent to the predictive module. Machine learning model classifies results produced by supervised learning methods [1] and returns and visualizes them to the stakeholders of the institution. This hierarchical flow of work provides effective communication between layers without modular integrity.



[Fig. System Architecture Diagram]

VI. IMPLEMENTATION

The Intelligent Alumni Management System proposed was deployed through the integration of both the web

development technologies and machine learning frameworks to make sure the seamless integration of both data management and predictive analytics. The system backend was established in the framework of Spring Boot, which allows creating applications quickly and organizing the communication in the form of REST. JSP was used to design the frontend interface together with HTML, CSS and JavaScript so as to allow an interactive user-friendly environment. The relational database management system that was used was MySQL to store the structured academic records, profiles of alumni members, job opportunities, event details and placement history.

The machine learning part has been created in Python and popular data science packages, such as NumPy, Pandas, and Scikit-learn [4]. It was cleaned and then processed into a dataset, which was used to train several supervised learning models [1]. Following a comparative analysis, the Random Forest classifier [3] has been adopted to be deployed because of the better predictive performance. The trained model was being serialized and it was stored in order to be re-used at runtime without having to be retrained.

Streamlined data exchange mechanisms were used to integrate the web application with the predictive model. Attributes of students, academic and skills, in the database are read and reconstructed as feature vectors that can be inputted in the trained model. In case of a request of an administrator or other authorized staff to conduct placement analysis, the system pulls out all pertinent data pertaining to the students and forwards it to the predictive module. The model produces a probability score that indicates a likelihood of placement, which is further transmitted back to the application level and displayed with the help of the administrative dashboard.

The implementation means that it has modular independence between the machine learning component and the web application. This decoupling allows updated predictive model to be made in the future without changes in the underlying alumni management features. Moreover, the system is scalable to process larger datasets and can be further extended to the cloud-based deployment environment in case it is vital.

By means of the implementation of predictive analytics to the alumni management platform, the implementation manages to turn conventional institutional data management into a smart decision-support mechanism that can predict the results of placement (in real-time).

VII. RESULTS

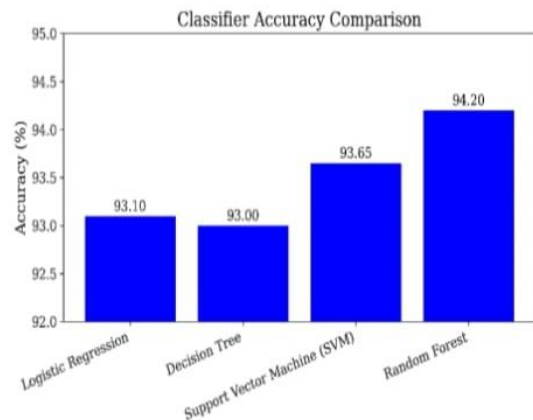
The effectiveness of predictive analytics in the prediction of the results of student placement can be proved by the experiments conducted to evaluate the machine learning models. Four supervised classification algorithms, including Decision Tree [2], Support Vector Machine [5], Logistic Regression [1], and Random Forest [3] were analyzed and compared in terms of measures of accuracy, precision, recall, F1-score, and ROC-AUC. As shown in the confusion matrices in Section IV, the classification performance of each of the models are shown with respect to true positives, true negatives, false positives, and false negatives.

The highest accuracy was found to be 94.20 and the highest ROC-AUC was 0.9868 with the Random Forest algorithm [3], which was the best classifier of the evaluated ones. This means that it has better discriminative ability in the separation of placed and non-placed students. Random Forest has an ensemble character which decreases the variance because it combines several decision trees thus minimizing the overfitting and enhancing the generalization performance [3]. The balance of its accuracy (0.9298) and recall (0.9320) is quite consistent as it identifies positive and negative placement outcomes.

The Support Vector Machine classifier [5] was also able to perform well with an accuracy of 93.65 and ROC-AUC of 0.9831. It has the capacity to maximize the separation of margins in the feature space which makes it give consistent classification [5]. The accuracy was 93.10, which was obtained with the help of the Logistic Regression [1], as it indicated that the linear correlations between academic characteristics have a great impact on the results of placing students. The Decision tree classifier [2] had an accuracy of 93.00% with interpretable decision limits, but with a little less robustness than the ensemble methods.

The comparison shows that the ensemble-based approaches [3] are better than individual classifiers in this prediction task. Although the accuracy of all models was above 93, Random Forest has always been on the top of all evaluation metrics. The findings confirm that academic performance, technical skills, internships, and certifications are the good predictive factors of placement success.

A Random Forest classifier [3] built into the Alumni Management System makes it possible to have a real-time placement prediction and promotes the proactive approach to academic intervention. Given the students who are under the threat of non-placement, the institution can develop specific training programs and skill improvement activities. The feasibility of the practical implementation and institutional relevance of integrating machine learning into alumni lifecycle management systems is thus validated by the outcomes of the experiment.



VIII. CONCLUSION

The study described the development and deployment of a Intelligent Alumni Management System with a Machine Learning-based Placement Prediction model. In comparison to the traditional alumni management systems, in which the main emphasis is made on the management of records and communication, the proposed system enriches the institutional capacities by having the element of predictive analytics directly incorporated into the management system. It is possible to activate the preparation of student placement knowledge in advance and thus make the system a data-driven decision-support tool through the integration of supervised learning techniques [1].

The 4 classification algorithms, namely Logistic Regression [1], Decision Tree [2], Support Vector Machine [5], and Random Forest [3] were implemented and compared with standard performance measures such as accuracy, precision, recall, F1-score, and ROC-AUC. The outcome of the experiments showed that all models were strong predictors with accuracy above 93. The best classification capability and robustness were measured as the highest accuracy value of 94.20 percent and the largest ROC-AUC value of 0.9868 belonging to the Random Forest classifier [3]. Random Forest was the final model to be deployed in the system because it has an ensemble learning mechanism, and its performance is balanced with both evaluation metrics [3].

The combination of the chosen predictive model into the Alumni Management System allows placing the probability of real-time estimation based on the academic and skill-related factors. This feature assists the administrators and the placement coordinators in creating specific intervention plans and strategies to be adopted by the students who need further academic or skill development. The system thus transforms the institutional placement management into a proactive and strategic planning mechanism as opposed to a reactive process mechanism.

IX. FUTURE WORK

Though the suggested Intelligent Alumni Management System combined with the placement prediction based on the use of the Machine Learning model proves to have the high predictive accuracy and high practicality, a few improvements can be addressed in further studies. It would enhance the generalization and strength of the models in various academic settings by increasing the size of the dataset by incorporating larger multi-institutional records. To increase predictive accuracy, interest in extracurricular activities, coding platform performance, behavioral indicators, aptitude assessment scores and real-time certifications of skills could be added attributes.

Further studies can also incorporate the use of higher-level deep learning structures, such as Artificial Neural Networks (ANN) and an ensemble methodology to learn more of the nuanced nonlinear connections in the placement data. The grid search and Bayesian optimization methods of

hyperparameter optimization can be used to optimize the model further. Also, by introducing explainable AI (XAI) techniques, these problems can be addressed and made more transparent by finding out that features matter and also offer interpretable insights into the placement probability to those in charge of the institution.

System-wise, the implementation of the framework within cloud-based environment would have increased the scalability and enabled it to integrate with institutional ERP systems. The predictive module can also be expanded to give personalized training advice to students upon the skill gaps identified. Moreover, the use of real-time analytics dashboard and automatic alerts systems might help administrators to monitor the placement readiness on batches by default.

The intelligent forecasting, which is integrated into the alumni lifecycle management, creates opportunities on the wider educational analytics use. The proposed framework can be modified into a structured institutional decision ecosystem through further developments in predictive modelling and the deployment of systems in the future.

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